

AML Transaction Monitoring Analysis

Project Overview:

This project analyses AML Transaction Monitoring across two source tables - a 2,000-row client master (static KYC/risk indicators) and 50,000-row transactions ledger (Behavioural/transaction-level indicator). The goal of this project is to build an **Anti-Money Laundering transaction monitoring and risk analytics system** that identifies suspicious activity, prioritizes high-risk clients, and presents the findings through clear reports and interactive visualizations.

Dataset Summary:

Table	Rows	Key columns
Client data	2,000	client_id, client_name, client_type, sector, sector_risk, country, pep_flag, sanctions_flag, fatf_country_flag, ofac_country_flag, sectoral_sanctions_flag, ownership_opacity_score
Transaction data	50,000	transaction_id, client_id, amount, transaction_type, timestamp, client_country, counterparty_country, ofac_match_flag, fatf_country_flag, structuring_pattern_flag, rapid_movement_flag, trade_mispricing_flag

Exploratory Data Analysis using Python:

We began with data preparation and data cleaning in Python.

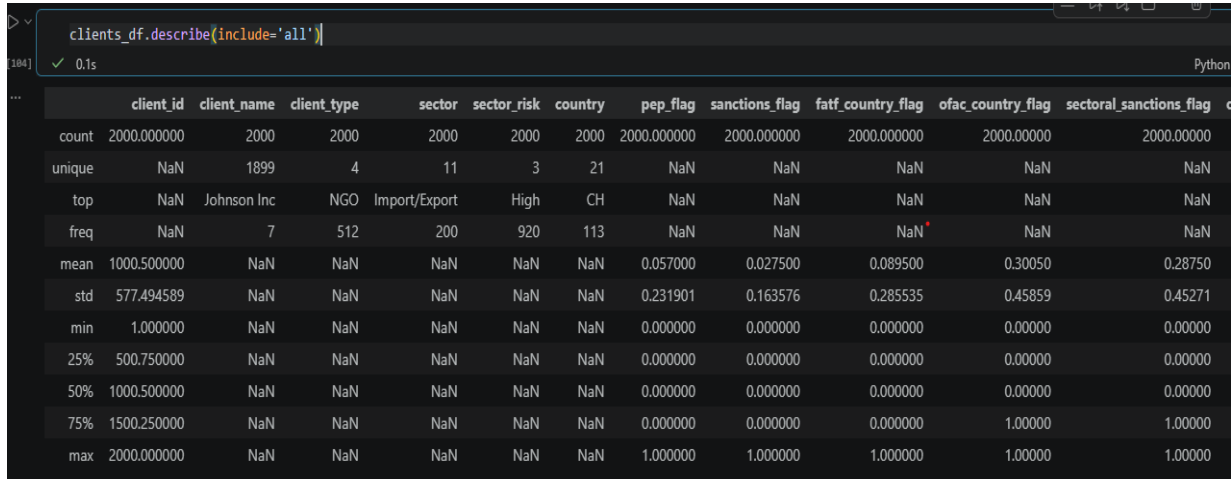
Data Loading:

Dataset is imported using Pandas and NumPy.

```
#Data Importing
import pandas as pd
import numpy as np
INPUT = "aml_transaction_data.xlsx"
clients_df = pd.read_excel(INPUT, sheet_name="client data", usecols="A:L", engine="openpyxl")
transaction_df = pd.read_excel(INPUT, sheet_name="transaction data")
```

Initial Exploration:

Used `df.info()` to check structure and `.describe()` for summary statistics.



```
clients_df.describe(include='all')
```

	client_id	client_name	client_type	sector	sector_risk	country	pep_flag	sanctions_flag	fatf_country_flag	ofac_country_flag	sectoral_sanctions_flag
count	2000.000000	2000	2000	2000	2000	2000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000
unique	NaN	1899	4	11	3	21	NaN	NaN	NaN	NaN	NaN
top	NaN	Johnson Inc	NGO	Import/Export	High	CH	NaN	NaN	NaN	NaN	NaN
freq	NaN	7	512	200	920	113	NaN	NaN	NaN	NaN	NaN
mean	1000.500000	NaN	NaN	NaN	NaN	NaN	0.057000	0.027500	0.089500	0.30050	0.28750
std	577.494589	NaN	NaN	NaN	NaN	NaN	0.231901	0.163576	0.285535	0.45859	0.45271
min	1.000000	NaN	NaN	NaN	NaN	NaN	0.000000	0.000000	0.000000	0.00000	0.00000
25%	500.750000	NaN	NaN	NaN	NaN	NaN	0.000000	0.000000	0.000000	0.00000	0.00000
50%	1000.500000	NaN	NaN	NaN	NaN	NaN	0.000000	0.000000	0.000000	0.00000	0.00000
75%	1500.250000	NaN	NaN	NaN	NaN	NaN	0.000000	0.000000	0.000000	1.00000	1.00000
max	2000.000000	NaN	NaN	NaN	NaN	NaN	1.000000	1.000000	1.000000	1.00000	1.00000

Data Cleaning:

Client Data: Every column is removed from Dataframe if it contains any NAN/NULL values. To ensure that Client IDs are consistent between the client and transaction datasets, `client_id` is converted to Numeric. While checking, it was found that clients data sheet had a corrupted used-range extending to 16,384 columns, filled with junk placeholder headers ("Column1" through "Column 16372") beyond the 12 real data columns. It is removed using built in module regular expression `re.match()` which checks for a letter or number at the very beginning and identify junk columns and `drop` is used to remove junk columns.



```
#Data Cleaning-clients
clients_df = clients_df.dropna(axis=1, how="all")
clients_df['client_id'] = pd.to_numeric(clients_df['client_id']).astype('Int64')
# Also drop columns with the junk placeholder header pattern ("Column1", "Column2", ...)
import re
junk_cols = [c for c in clients_df.columns if re.match(r"^\d+Column\d+$", str(c))]
clients_df = clients_df.drop(columns=junk_cols)

print(f"Remaining columns: {len(clients_df.columns)}")
print(clients_df.columns.tolist())
```

Remaining columns: 12
['client_id', 'client_name', 'client_type', 'sector', 'sector_risk', 'country', 'pep_flag', 'sanctions_flag', 'fatf_country_flag', 'ofac_country_flag', 'sectoral_sanctions_flag']

Transactions Data: Dropped fully-blank rows and enforced numeric/text types on required column. Trimmed whitespace, standardized country codes and transaction type to uppercase. Transaction timestamps are stored as Excel serial date numbers which is an incorrect unit parameter (defaulting to nanoseconds) collapsed every date to 1970-01-01. The correct method to convert to real datetime as below is used.

```
#Data cleaning-Transaction
transaction_df["transaction_type"] = transaction_df["transaction_type"].str.upper()
transaction_df["timestamp"] = pd.to_datetime(transaction_df["timestamp"], unit="D", origin="1899-12-30")
transaction_df["transaction_date"] = transaction_df["timestamp"].dt.date
transaction_df["transaction_month"] = transaction_df["timestamp"].dt.to_period("M").astype(str)
transaction_df["transaction_hour"] = transaction_df["timestamp"].dt.hour
transaction_df["timestamp_display"] = transaction_df["timestamp"].dt.strftime("%Y-%m-%d %H:%M:%S")
transaction_df['amount'] = pd.to_numeric(transaction_df['amount'])
transaction_df['client_id'] = pd.to_numeric(transaction_df['client_id']).astype('Int64')
client_flag_cols = ['pep_flag', 'sanctions_flag', 'fatf_country_flag', 'ofac_country_flag', 'sectoral_sanctions_flag', 'ownership_opacity_score']
for col in client_flag_cols:
    clients_df[col] = pd.to_numeric(clients_df[col]).fillna(0).astype(int)
txn_flag_cols = ['ofac_match_flag', 'fatf_country_flag', 'structuring_pattern_flag', 'rapid_movement_flag', 'trade_mispricing_flag']
for col in txn_flag_cols:
    transaction_df[col] = pd.to_numeric(transaction_df[col]).fillna(0).astype(int)
```

✓ 0.1s

Data Merging:

Every transaction to the corresponding client profile is connected through Data Merging.

```
#Data Merging
analysis_df = transaction_df.merge(clients_df, on="client_id", how="left", validate="many_to_one", suffixes=("_txn", "_client"))
```

✓ 0.1s

The merge produced:

- 50000 Transaction records
- 2000 Unique Clients
- 0 Unmatched clients

Feature Engineering:

Client-Level Risk calculation:

Here client-level AML risk is calculated and then share of transactions belonging to high-risk clients is determined.

The columns that represent **client-level risk factors** are

- ❖ pep_flag → PEP indicator
- ❖ sanctions_flag → sanctions indicator
- ❖ fatf_country_flag_client → FATF jurisdiction indicator
- ❖ ofac_country_flag → OFAC-related country indicator
- ❖ sectoral_sanctions_flag → sectoral sanctions indicator
- ❖ ownership_opacity_score → ownership transparency risk

The total amount of client-level risk indicated by these columns is calculated. A client is high-risk if sector risk is High OR the client has at least one risk indicator.

```
#Client level risk calculation
client_risk_flag_cols = ['pep_flag', 'sanctions_flag', 'fatf_country_flag_client', 'ofac_country_flag', 'sectoral_sanctions_flag', 'ownership_opacity_score']
analysis_df['client_flag_count'] = analysis_df[client_risk_flag_cols].sum(axis=1)
analysis_df['high_risk_client'] = ((analysis_df['sector_risk'] == "High") |
    (analysis_df['client_flag_count'] > 0)).astype(int)
analysis_df['high_risk_client_transaction_share'] = (
    analysis_df['high_risk_client'].sum()
    / analysis_df['transaction_id'].count()
)
```

✓ 1.8s Python

Transaction-Level alert calculation:

Transaction-level AML risk calculation identifies whether each transaction is suspicious based on multiple transaction-monitoring rules, then calculates overall suspicious transaction KPIs.

Each column represents a different AML rule:

Flag	Meaning
ofac_match_flag	Transaction has an OFAC-related match
fatf_country_flag_txn	Transaction involves a FATF high-risk jurisdiction
structuring_pattern_flag	Possible structuring/smurfing pattern
rapid_movement_flag	Money moves rapidly through the account
trade_mispricing_flag	Possible trade mispricing

The transaction-level risk rules fired for each transaction is calculated. If at least one transaction risk rule fired, the transaction is classified as suspicious. The number of transactions associated with each client is calculated and hence suspicious transaction rate is determined. The client-level risk **and** transaction-level KPI are being connected through the merged analysis_df.

```
#transaction level risk calculation
txn_risk_flag_cols = ['ofac_match_flag', 'fatf_country_flag_txn', 'structuring_pattern_flag', 'rapid_movement_flag', 'trade_mispricing_flag']
analysis_df['txn_flag_count'] = analysis_df[txn_risk_flag_cols].sum(axis=1)
analysis_df['suspicious_txn'] = (analysis_df['txn_flag_count'] > 0).astype(int)
analysis_df['txn_per_client'] = analysis_df.groupby("client_id")["client_id"].transform("count")
analysis_df['suspicious_transaction_count'] = analysis_df['suspicious_txn'].sum()
analysis_df['suspicious_transaction_rate'] = (
    analysis_df['suspicious_txn'].sum() / analysis_df['transaction_id'].count()
) * 100
```

✓ 0.0s

Risk Score Calculation:

A Composite AML risk score is calculated by taking four different risk components, converting them into numerical scores, adding them together, and then classifying the result as **Low, Medium, or High**.

Later Overall risk rating is classified based on below rule.

Composite Score	Risk Rating
5 or more	High
3–4	Medium
0–2	Low

Data Integration:

Connected Python Script to MYSQL. A database named “transaction_analysis” is created in MYSQL and loaded the cleaned data frame into the database for SQL analysis.

Data Analysis using SQL:

Tables “mergeddata” and “risk rating” has one row per transaction with client attributes already joined in. All queries below were tested against this dataset to derive some key findings

```
1 #Database creation and data imported from python
2 • create database transaction_analysis
3 ✖ use transaction_analysis
4 describe mergeddata
5 select * from mergeddata
```

	transaction_id	client_id	amount	transaction_type	timestamp	client_country	counterparty_country	ofac_match_flag	fatf_country_flag_txn
▶	1	1412	1027.64	CHECK	2025-08-20 09:21:56	IN	SG	0	0
	2	1035	518.82	SWIFT	2025-09-07 07:52:58	RU	NL	0	0
	3	704	460.78	SWIFT	2025-09-28 03:00:43	LB	IN	0	0
	4	1073	1625.76	WIRE	2025-08-22 04:57:11	AE	CA	0	0
	5	1622	1856.47	SWIFT	2025-08-07 19:49:50	CH	IN	0	0
	6	1053	2786.86	SWIFT	2025-08-25 22:09:24	IR	US	0	0
	7	999	513.52	SWIFT	2025-09-06 11:54:17	CH	JP	0	0
	8	681	1884.2	WIRE	2025-08-02 20:42:22	NL	CN	0	0
	9	637	325.81	SWIFT	2025-09-08 05:11:54	US	UK	0	0
	10	578	322.84	ACH	2025-08-13 17:55:46	FR	CH	0	0
	11	1541	15.27	WIRE	2025-08-15 04:28:33	AU	CA	0	0
	12	283	2712.83	WIRE	2025-09-03 20:08:57	IN	CN	0	0
	13	964	2352.74	WIRE	2025-08-26 21:31:33	KP	IN	0	0
	14	1203	2177.45	ACH	2025-09-12 08:03:00	NL	DE	0	0
	15	279	2574.84	ACH	2025-07-16 15:40:49	VE	NL	0	0
	16	536	1326.71	CHECK	2025-09-23 18:56:39	US	NL	0	0
	17	204	2070.57	WIRE	2025-07-27 00:22:20	IN	AE	0	0

#DATA QUALITY ANALYSIS:

To know the dataset is appropriate for further analysis, data quality checks has been done.

```
8 -- Q1.Does every transaction have a matching client?
9 SELECT COUNT(*) AS orphaned_transactions FROM mergeddata WHERE client_name IS NULL;
10
```

orphaned_transactions
0

```
11 -- Q2.Are there duplicate transaction_ids that would double-count volume?
12 • SELECT transaction_id, COUNT(*) AS dup_txn FROM mergeddata GROUP BY transaction_id
13 HAVING COUNT(*) > 1;
14
```

transaction_id	dup_txn
----------------	---------

```

15    -- Q3.Any transactions with null/zero/negative amounts that shouldn't be in the "clean" dataset?
16 •   SELECT COUNT(*) AS bad_amount_rows FROM mergeddata WHERE amount IS NULL OR amount <= 0;
17

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	bad_amount_rows			
	0			

#PORTFOLIO ANALYSIS

Key Performance Indicators:

```

19    -- Q4.What's the overall shape of the book – volume, value, and how much of it is flagged?

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
kpi_name	kpi_value			
Total Transactions	50000			
Total Transaction Amount	209806892.97			
Average Transaction Amount	4196.14			
Unique Customers	2000			
Suspicious Transactions	4489			
Suspicious Transaction Rate (%)	8.978			
Suspicious Transaction Amount	22218264.360000003			
PEP-Flagged Client Transactions	2887			
Sanctioned Client Transactions	1419			
OFAC Match Count	1790			
FATF Country Flag Count (Txn)	602			
Structuring Pattern Count	347			
Rapid Movement Count	2460			
Trade Mispricing Count	10			

- Total Transactions counts to **50 k** and unique customers found to be **2000**.
- Total transaction value of approximately **\$209.8 million**
- **4,489 suspicious transactions**, representing **8.98%** of all transactions
- Suspicious transactions represented approximately **\$22.2 million**, or **10.6%** of total transaction value.
- **Rapid movement** was the dominant alert driver with **2,460 flagged transactions** and **OFAC match** accounted for **1,790 flagged transactions** which is **2nd highest**.

Unique Customers:

52 -- Q5 .Find the list of unique customers ?

client_id	client_name	client_type	sector	country	transaction_per_client	transaction_amount	Suspicious_txn_count	dnt
1	Wells-Turner	Financial Institution	NGO/Charity	JP	26	261181.94	1	0
2	Goodman, Hendrix and Garcia	Financial Institution	Import/Export	CH	28	59430.99	0	0
3	Phillips-Hanson	NGO	NGO/Charity	AE	25	61924.06	22	25
4	Kim, Anderson and Espinoza	Financial Institution	Defense/Arms	RU	29	210226.5	3	87
5	Marks-Thompson	Financial Institution	Financial Services	AU	22	61420.28	0	22
6	Jones Inc	Corporate	NGO/Charity	AE	23	181771.64	3	23
7	Lambert-Ruiz	Individual	Energy/Oil	US	26	85070.08	2	26
8	Thompson Inc	NGO	Consulting	CA	26	43633.21	3	0
9	Vega-Randall	NGO	Financial Services	US	17	38554.41	0	34
10	Edwards PLC	Corporate	Casino/Gambling	SY	25	327501.73	0	25
11	Solis-Kemp	NGO	NGO/Charity	AF	22	242607.88	3	0
12	Stein-Davis	NGO	Import/Export	SD	20	35571.72	1	20
13	Gonzalez-Jenkins	Financial Institution	Retail	NL	26	46490.71	1	0
14	Harris-Walker	Individual	Real Estate	RU	31	63146.91	2	31
15	Adkins, Howard and Mcdonald	Corporate	Energy/Oil	SG	25	131889.93	3	25
16	Estrada Ltd	Financial Institution	Energy/Oil	IR	21	194140.84	2	63
17	Stein, Webb and Smith	Financial Institution	Defense/Arms	NL	33	187925.78	1	33
18	Christian, Shepard and Wells	Financial Institution	Tech	AE	24	80256.21	0	24

The unique-client analysis shows substantial variation in transaction volume, transaction value, and suspicious activity across clients. Phillips-Hanson stands out with 22 suspicious transactions out of 25, while several high-value clients show no suspicious transactions. This indicates that transaction value and transaction volume alone do not explain suspicious activity; client-specific transaction behaviour and AML alert patterns should also be considered when prioritizing investigations.

Top 20 clients:

91 -- Q6. Find the top 20 clients who has highest transaction value?

client_id	client_name	Total_transaction_value	pct_of_total
371	Hudson-Bell	543808.47	0.26
1551	Reid-Peck	521373.07	0.25
1015	Nguyen-Parsons	520590	0.25
1245	Camacho Group	493986.81	0.24
1934	Coffey, Hudson and Gomez	492033.69	0.23
1026	Liu-Hurley	479716.04	0.23
1179	Thomas, Flores and Hansen	477127.47	0.23
1140	Stokes, Ashley and Gutierrez	466791.62	0.22
779	Alvarado, Taylor and Black	465545.74	0.22
1805	Nunez, Roy and White	444026.85	0.21
1512	Jones-Kramer	441469.27	0.21
539	Moore Inc	432252.79	0.21
710	Walter and Sons	412080.43	0.2
845	Brown Inc	409302.02	0.2
1476	Martinez-Shaw	389081.53	0.19
906	Cole LLC	386869.3	0.18
237	Davis Ltd	386305.1	0.18
677	Hayes and Sons	377764.32	0.18
1283	Fowler, Smith and Bishop	377581.33	0.18

Result 133

The top 20 clients by transaction value show relatively distributed transaction exposure, with Hudson-Bell recording the highest total value at approximately \$ 543.8 K. The leading clients individually account for only around 0.18%–0.26% of total transaction value, indicating that overall transaction value is not heavily concentrated in a single client. High transaction value should therefore be assessed alongside suspicious activity, client risk indicators, and behavioural patterns when prioritizing AML monitoring

#CLIENT RISK SEGMENTATION

Classification of Clients on risk category:

102 -- Q7. How many clients fall into each risk rating, and how much money moves through each tier?

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	overall_risk_rating	client_count	total_value	suspicious_rate_pct
▶	High	1218	159690663.04	11.60
	Medium	634	43866640.99	4.99
	Low	148	6249588.94	1.53

```
risk_summary = (  
    client_risk_df["overall_risk_rating"]  
    .value_counts()  
    .rename_axis("overall_risk_rating")  
    .reset_index(name="client_count")  
)  
  
print(risk_summary)
```

	overall_risk_rating	client_count
0	High	1218
1	Medium	634
2	Low	148

High-risk clients (1218) represent 60.9% of the client population and account for approximately \$1.597 billion, or 78.2% of total transaction value. They also have the highest suspicious transaction rate at 11.60%, compared with 4.99% for Medium-risk and 1.53% for Low-risk clients. The concentration of both client population and transaction value within the High-risk tier makes this segment a major focus for AML monitoring. The clear decrease in suspicious rates from High to Medium to Low also indicates that the risk-rating framework is broadly aligned with observed transaction-level suspicious activity.

Top Clients across Multiflag Analysis:

168 -- Q8. Find the top 20 clients who carry multiple static risk flags simultaneously?

client_id	client_name	sector	country	pep_flag	sanctions_1	fatf_country	ofac_country	sectoral_sanct	ownership_opacit	client_flag_count
886	Rivera, Ross and Smith	Energy/Oil	KP	1	0	1	1	1	1	5
97	Powell-Johnson	Energy/Oil	SY	0	1	0	1	1	1	4
254	Saunders-Contreras	Energy/Oil	KP	0	1	1	1	1	0	4
333	Mendoza Ltd	Energy/Oil	IR	0	1	1	1	1	0	4
350	Miles, Schmidt and Par...	Defense/Arms	IR	0	0	1	1	1	1	4
492	Garcia-Miller	Financial Ser...	VE	1	0	0	1	1	1	4
583	Smith, Powers and Col...	Financial Ser...	KP	0	0	1	1	1	1	4
933	Brown, Smith and Guti...	Defense/Arms	KP	0	0	1	1	1	1	4
1161	Nicholson-Harris	Financial Ser...	IR	1	0	1	1	1	0	4
1395	May-King	Energy/Oil	IR	1	0	1	1	1	0	4
1503	Ward-Harris	Financial Ser...	VE	1	1	0	1	1	0	4
1819	Hamilton-Griffin	Energy/Oil	RU	1	0	0	1	1	1	4
1875	Choi, Welch and Mitchell	Financial Ser...	IR	1	0	1	1	1	0	4
4	Kim, Anderson and Es...	Defense/Arms	RU	1	0	0	1	1	0	3
16	Estrada Ltd	Energy/Oil	IR	0	0	1	1	1	0	3
30	Moore, Ballard and Ad...	Casino/Gambl...	IR	0	0	1	1	0	1	3
45	Ryan, Lawson and Ha...	Financial Ser...	KP	0	0	1	1	1	0	3
48	Hill-Luna	Financial Ser...	KP	0	0	1	1	1	0	3
53	Johnson LLC	Defense/Arms	IR	0	0	1	1	1	0	3
82	Walker-Norman	Energy/Oil	VE	1	0	0	1	1	0	3

The multi-flag analysis identifies several clients with four or more simultaneous static AML/KYC risk indicators. The strongest combinations are primarily driven by FATF-country, OFAC-country and sectoral-sanctions indicators, with PEP and ownership-opacity flags adding further risk dimensions for selected clients. Multi-flag clients are distributed across several sectors, with Energy/Oil and Defence/Arms appearing repeatedly. These clients represent a concentrated population of static-risk indicators and can be prioritized for enhanced KYC review and validation of beneficial ownership and sanctions exposure

PEP vs NON-PEP clients:

177 -- Q9. Do PEP clients actually transact differently than non-PEP clients, or is the flag not predictive?

pep_status	clients	avg_txn_amount	suspicious_rate_pct
Non-PEP	1886	4063.21	8.87
PEP	114	6365.42	10.81

- PEP clients transact at significantly higher value-The average transaction amount for PEP clients is 6,365.42, compared with 4,063.21 for non-PEP clients which is about 57% higher.
- PEP clients have a higher suspicious transaction rate-The suspicious rate for PEP clients is 10.81% vs 8.87% for non-PEPs. This is a 1.94 percentage-point increase, or roughly 22% higher relative to non-PEPs.
- PEP status appears to be associated with elevated transaction risk. The combination of higher transaction values + higher suspicious rates suggests that PEP clients deserve enhanced monitoring.

Clients with High Suspicious Rate:

186 -- Q10. Find Clients with very high suspicious-transaction rate?

client_id	client_name	total_txns	suspicious_txns	suspicious_rate_pct
822	Snyder PLC	12	12	100.0
1785	Alexander-Johnson	30	28	93.3
714	Gallegos and Sons	25	22	88.0
3	Phillips-Hanson	25	22	88.0
333	Mendoza Ltd	24	21	87.5
731	Davis Inc	22	19	86.4
768	Thomas-Martin	29	25	86.2
1879	Dixon-Butler	28	24	85.7
97	Powell-Johnson	21	18	85.7
1632	Griffin-Howard	20	17	85.0
1323	Gregory, Fisher and J...	33	28	84.8
1216	Jackson, Smith and Ri...	26	22	84.6
403	Li Ltd	25	21	84.0
944	Suarez Group	25	21	84.0
1215	Barton, Davis and Tur...	23	19	82.6
1788	Merritt LLC	28	23	82.1
1754	Green, Burns and Ward	28	23	82.1
1433	Lozano, Oconnell and ...	32	26	81.3
1779	Hawkins Ltd	26	21	80.8
1027	Cowan-Howe	20	16	80.0

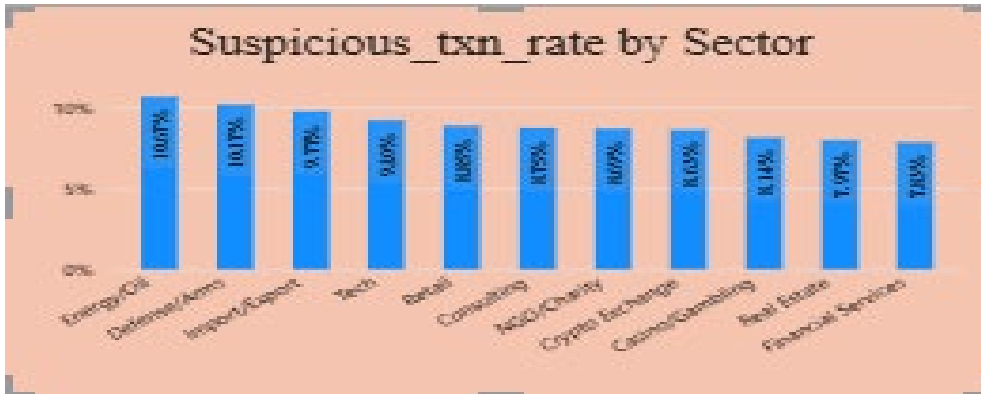
The analysis identifies a group of clients where more than 80% of their transactions are flagged as suspicious. Snyder PLC is particularly notable because all 12 transactions are flagged. Several other clients have both high transaction volumes and suspicious rates above 80%, suggesting potential concentrations of transaction-monitoring alerts that warrant further investigation.

#SECTOR ANALYSIS:

Sector with highest suspicious rate:

199 -- Q11. Which sectors carry the highest suspicious-transaction rate?
200 -- where should enhanced due diligence be concentrated?

sector	clients	transactions	total_value	suspicious_rate_pct
Energy/Oil	187	4735	29790211.85	10.67
Defense/Arms	192	4751	30645600.2	10.17
Import/Export	200	4993	11746739.24	9.77
Tech	154	3847	9187493.31	9.20
Retail	172	4224	9791318.8	8.85
Consulting	169	4206	8939024.76	8.75
NGO/Charity	188	4762	30221431.38	8.69
Crypto Excha...	173	4382	28154997.91	8.63
Casino/Gambl...	180	4556	28059975.32	8.14
Real Estate	189	4706	11469709.92	7.97
Financial Ser...	196	4838	11800390.28	7.83



Energy/Oil has the highest suspicious-transaction rate at 10.67%, followed by Defence/Arms at 10.17% and Import/Export at 9.77%. These sectors show comparatively higher concentrations of transactions triggering AML monitoring rules and can be examined more closely for the specific alert types driving the elevated rates.

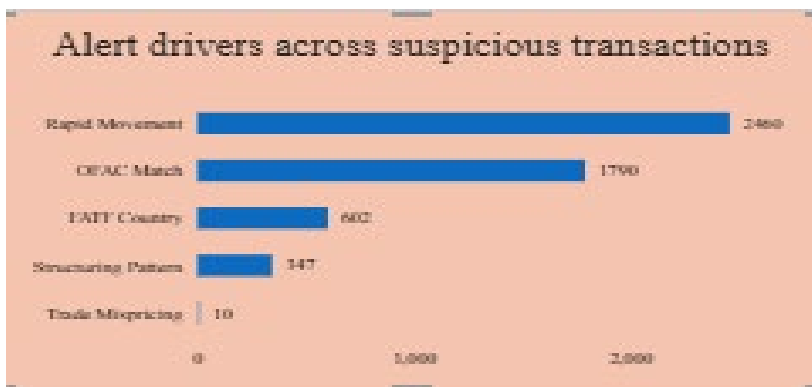
Sector specific Alert Drivers:

211 -- Q12.Within each sector, which specific alert type fires most ?

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Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	sector	ofac_match	fatf_country	structuring	rapid_movement	trade_mispricing
▶	Casino/Gambl...	114	36	49	215	0
	Consulting	160	46	17	203	0
	Crypto Excha...	129	37	45	211	0
	Defense/Arms	212	81	46	236	0
	Energy/Oil	237	88	33	256	0
	Financial Ser...	108	32	27	250	0
	Import/Export	199	64	26	273	10
	NGO/Charity	169	53	42	211	0
	Real Estate	130	44	23	232	0
	Retail	169	50	20	197	0
	Tech	163	71	19	176	0



- **Rapid movement is highest** in every single sector with overall 2460 flagged transactions which contradicts that rapid movement might be concentrated in Crypto Exchange and Financial Services due to their legitimately high-velocity business models.
- **Casinos have strong structuring venues** (cash-intensive, chip purchases just under reporting thresholds), and crypto exchanges are the modern equivalent. The rule concentrating in exactly those two is strong evidence it's working as intended — this is the kind of domain-consistent pattern that suggests real signal rather than noise.
- **Energy/Oil** (237 + 88) and **defence/Arms** (212 + 81) lead in absolute terms, which is unsurprising given how heavily sanctioned oil and arms trade are. But **Tech** with 71 FATF hits third-highest in absolute terms despite being a mid-sized sector, and the **highest sanctions-driven share of any sector (54.5%)**. Given export controls on semiconductors and dual-use technology, that's worth a specific look rather than being lumped in with lower-risk sectors.

Sector risk vs suspicious rate:

223 -- Q13: Does sector_risk actually line up with the observed suspicious rate,
224 -- or are some "Low" sectors quietly running hot?

sector	sector_risk	observed_suspicious_rate_pct
Energy/Oil	High	10.67
Defense/Arms	High	10.17
Import/Export	Medium	9.77
Tech	Low	9.20
Retail	Low	8.85
Consulting	Low	8.75
NGO/Charity	High	8.69
Crypto Excha...	High	8.63
Casino/Gambl...	High	8.14
Real Estate	Medium	7.97
Financial Ser...	Medium	7.83

Energy/Oil and Defence/Arms, both classified as High sector risk, recorded the highest suspicious transaction rates at 10.67% and 10.17%, respectively. This combination of elevated inherent sector risk and higher alert activity indicates a need for closer transaction monitoring and investigation of the underlying alert patterns, clients, and transactions. Although, the Technology sector is classified as Low Risk at the sector level, it has a suspicious transaction rate of 9.20%. This indicates that transaction-level monitoring is identifying potentially unusual activity within the sector. Further investigation is required to determine whether the alerts are concentrated among specific clients, transaction types, countries, or AML detection rules. Real Estate and Financial Services, both classified as medium sector risk, recorded the lowest suspicious transaction rates at 7.97% and 7.83%, respectively. This indicates comparatively lower observed alert activity than the other sectors analysed, although continued monitoring is warranted because both sectors still show a measurable level of suspicious transactions.

#COUNTRY ANALYSIS:

Top Client Countries with highest suspicious transaction rate and flag density:

233 -- Q14. Which client countries carry the highest suspicious rate and flag density?

	country	clients	transactions	suspicious_rate_pct	fatf_flagged_clients	ofac_flagged_clients
▶	VE	108	2722	12.71	0	108
	US	92	2261	11.50	0	0
	SY	105	2662	11.16	0	105
	LB	80	2009	10.55	0	0
	CH	113	2718	10.49	0	0
	AE	85	2182	9.72	0	0
	RU	99	2421	9.54	0	99
	SG	97	2421	8.96	0	0
	KP	84	2143	8.87	84	84
	AU	97	2471	8.86	0	0
	JP	88	2237	8.63	0	0
	CA	101	2561	8.00	0	0
	IN	94	2287	8.00	0	0
	FR	105	2640	7.95	0	0
	CN	83	2041	7.94	0	0
	SD	110	2725	7.82	0	110
	NL	96	2337	7.70	0	0
	AF	95	2427	7.66	0	0
	IR	95	2329	7.56	95	95
	UK	82	2153	7.29	0	0

Venezuela (VE), had the highest suspicious transaction rate among client countries with 12.71%. Venezuela has 108 clients, with all 108 classified as OFAC-flagged, resulting in a 100% OFAC-flag rate. This indicates a significant concentration of sanctions-related risk within the Venezuelan client population and highlights the need for enhanced KYC and sanctions screening. North Korea (KP) and Iran (IR) show 100% FATF and OFAC flag rates, meaning all clients associated with these countries are flagged by both country-risk indicators. This represents a concentrated sanctions and country-risk exposure and highlights the need for enhanced KYC, sanctions screening, and transaction monitoring. The underlying country-flag logic should be validated to distinguish country-level risk classification from individual sanctions-list matches.

Cross-country corridor analysis:

```
247 -- Q15: Cross-border corridor analysis – which client_country, counterparty_country PAIRS carry
248 -- the most flagged activity? (a single country view misses corridor-specific risk,
249 -- e.g. a "safe" country sending heavily to a high-risk one)
250
```

client_country	counterparty_country	transactions	total_value	suspicious_transactions	suspicious_rate_pct
SY	RU	36	117597.1	36	100.00
VE	IR	34	196536.79	34	100.00
VE	RU	34	116108.41	34	100.00
SY	SY	32	127055.64	32	100.00
VE	SY	31	100343.96	31	100.00
VE	VE	27	195768	27	100.00
SY	IR	27	227229.39	27	100.00
SY	VE	26	139089.68	26	100.00
VE	KP	26	316844.45	26	100.00
VE	SD	25	270113.96	25	100.00
SY	SD	25	81698.31	25	100.00
US	VE	25	355598.48	25	100.00
US	SY	23	121439.27	23	100.00
US	SD	23	95107.62	23	100.00
CH	VE	23	185537.33	23	100.00
RU	SD	22	127275.85	22	100.00
RU	VE	22	80055.27	22	100.00
AU	IR	22	123956.87	22	100.00
US	IR	21	156233.89	21	100.00
CH	SD	21	99579.33	21	100.00

The corridor analysis shows concentrated transaction-monitoring alerts, with most corridors recording a 100% suspicious transaction rate. Corridors involving Venezuela, Syria, Russia and Iran appear repeatedly, indicating concentrated country-pair exposure. The consistently high alert rate warrants further investigation of the underlying transaction patterns and validation of the alert rules to determine whether country-risk flags are driving the results.

Cross-border Vs Domestic transactions in country Analysis:

```
262 -- Q16: Cross-border vs domestic – do international transactions carry
263 -- materially more risk than same-country ones?
```

txn_scope	transactions	total_value	suspicious_transactions	suspicious_rate_pct
Cross-Border	47581	199706828.96	4247	8.93
Domestic	2419	10100064.01	242	10.00

Cross-border transactions account for the majority of transaction activity and value, with 47,581 transactions and approximately 199.7 M in total value. However, domestic transactions show a higher suspicious transaction rate of 10.00%, compared with 8.93% for cross-border transactions. This indicates that domestic transactions have a slightly higher observed alert

frequency, while cross-border transactions generate a substantially larger number of suspicious transactions due to their much higher volume

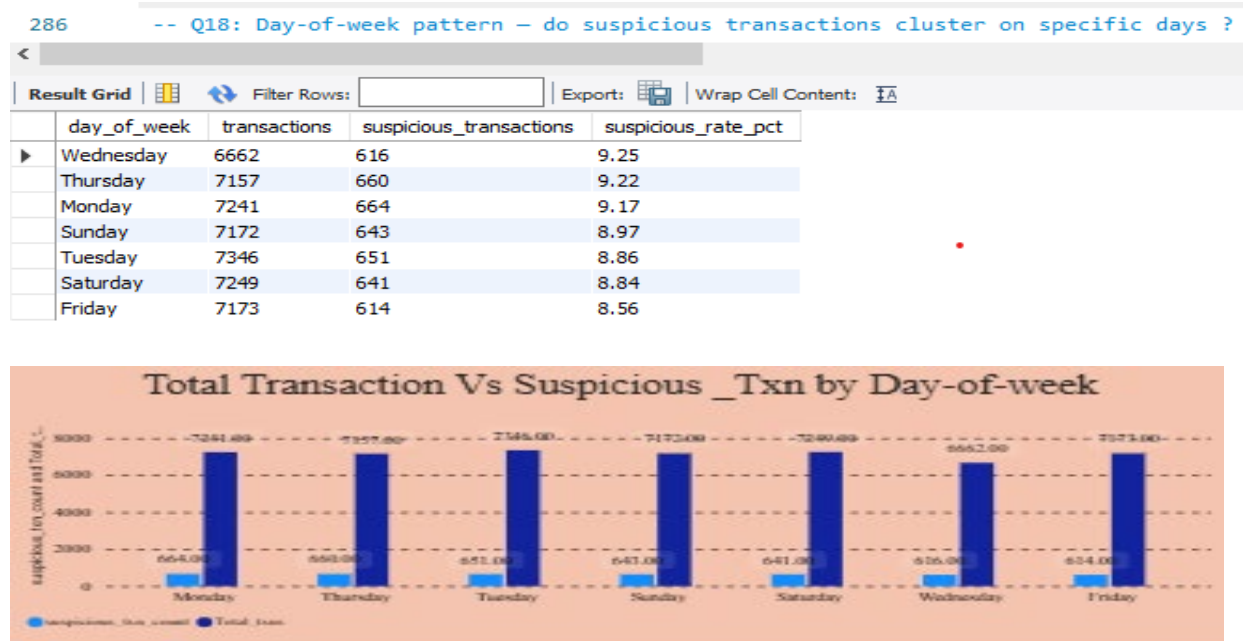
#TREND ANALYSIS

Suspicious Activity over Month:



From July to September 2025, transaction volume increased from 16,141 to 17,260, while the suspicious transaction rate gradually declined from 9.19% to 8.81%. Despite the lower rate, the absolute number of suspicious transactions increased to 1,520 in September due to higher transaction volume. Transaction value peaked at \$76.01 M in August before declining to \$70.86 M in September. Overall, the data shows growing transaction activity alongside a modest decline in the proportion of transactions flagged as suspicious; additional investigation outcomes would be required to determine whether this represents improved monitoring effectiveness

Suspicious Activity on day-to-day Basis:

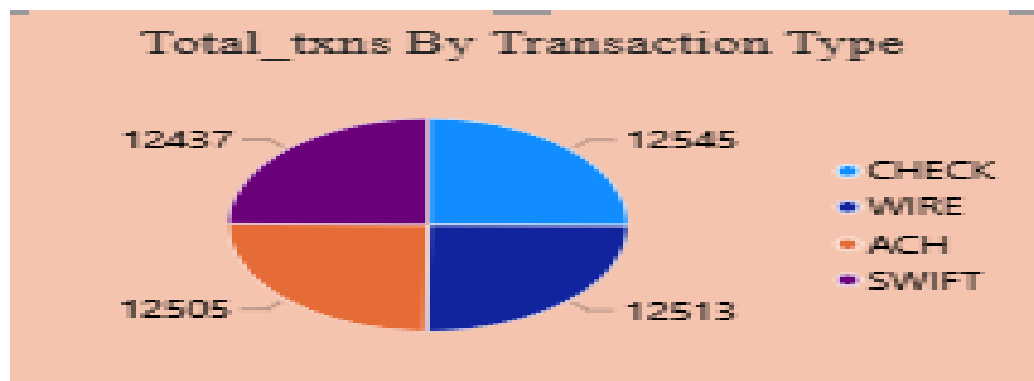


Suspicious transaction rates are relatively consistent across the week, ranging from 8.56% on Friday to 9.25% on Wednesday. Wednesday records the highest suspicious rate, while Monday generates the highest absolute number of suspicious transactions at 664. The relatively narrow 0.69 percentage-point range suggests that suspicious activity is broadly distributed across the week rather than being strongly concentrated on a particular day

Suspicious activity Across Transaction Type:

306 -- Q19:Which Transaction_type(Wire, SWIFT, ACH,Check) has bigger share of flagged activity?

transaction_type	transactions	total_value	suspicious_transactions	suspicious_rate_pct
CHECK	12545	54140411.8	1172	9.34
WIRE	12513	51602925.87	1142	9.13
SWIFT	12437	50740685.31	1100	8.84
ACH	12505	53322869.99	1075	8.60



CHECK transactions recorded the highest suspicious transaction rate at 9.34%, followed by WIRE at 9.13%, SWIFT at 8.84%, and ACH at 8.60%. Although CHECK has the highest observed rate, the difference between transaction types is relatively small at 0.74 percentage points, indicating that suspicious activity is broadly distributed across transaction channels rather than concentrated in a single transaction type

#ALERT DRIVERS ANALYSIS:

311 -- Q20:Which individual alert rule fires most – where's the monitoring workload actually concentrated?

flag_name	flag_level	records_triggered	population_base	triggered_rate_pct
rapid_movement_flag	Transaction	2460	50000	4.92
ofac_match_flag	Transaction	1790	50000	3.58
fatf_country_flag	Transaction	602	50000	1.20
structuring_pattern_flag	Transaction	347	50000	0.69
trade_mispricing_flag	Transaction	10	50000	0.02
ofac_country_flag	Client	601	2000	30.05
sectoral_sanctions_flag	Client	575	2000	28.75
fatf_country_flag	Client	179	2000	8.95
ownership_opacity_score	Client	120	2000	6.00
pep_flag	Client	114	2000	5.70
sanctions_flag	Client	55	2000	2.75

The alert-rule analysis shows that monitoring workload is concentrated primarily in Rapid Movement (2460) and OFAC Match (1790) at the transaction level, which together account for approximately 81.6% of transaction-level rule triggers. At the client level, OFAC Country (601) and Sectoral Sanction (575) have the highest trigger rates at 30.05% and 28.75%, respectively. In contrast, Trade Mispricing generates very few alerts. These results indicate that transaction monitoring workload is concentrated in a small number of rules, while client-level risk identification is largely driven by country and sanctions-related indicators. Alert volume should be evaluated alongside investigation outcomes and false-positive rates to assess rule effectiveness.

#TRANSACTION PATTERN/STRUCTURING PATTERN ANALYSIS

Effect of structuring pattern on rapid movement:

359 -- Q21: Rule co-occurrence – when structuring_pattern_flag fires, does rapid_movement_flag tend to fire alongside it?

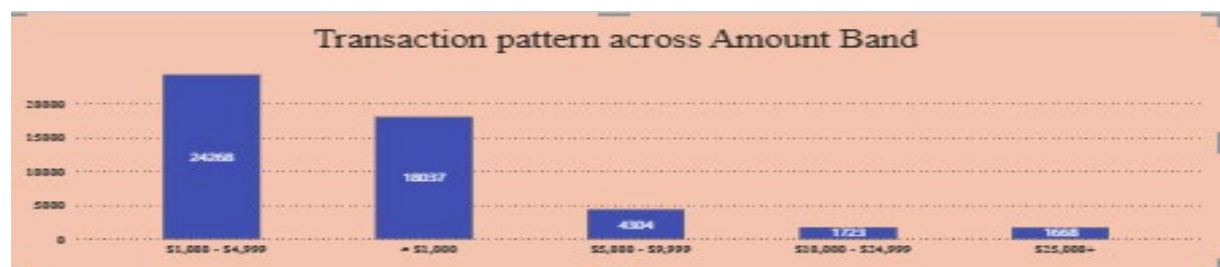
	total_structuring_alerts	also_rapid_movement	co_occurrence_pct
▶ 347	11	3.2	

Of the 347 transactions flagged for structuring patterns, only 11 also triggered the rapid movement rule, resulting in a 3.2% co-occurrence rate. This indicates that the two rules have limited overlap and are generally detecting different transaction patterns, suggesting complementary coverage within the transaction-monitoring framework.

Transaction Amount Band Analysis:

357 -- Q22. Find the Distribution of transaction sizes to see where volume concentrates?

	amount_band	transaction_count	suspicious_trar	suspicious_rate_pct
▶	< \$1,000	18037	1536	8.52
	\$1,000 - \$4,999	24268	1961	8.08
	\$5,000 - \$9,999	4304	664	15.43
	\$10,000 - \$24,999	1723	159	9.23
	\$25,000+	1668	169	10.13



Transactions in the \$5,000–\$9,999 range recorded the highest suspicious transaction rate at 15.43%, significantly exceeding the other transaction bands. Meanwhile, the \$1,000–\$4,999 range had the highest transaction volume and generated the largest number of suspicious transactions at 1,961. Transactions above \$25,000 also showed an elevated suspicious rate of 10.13%. The results indicate that suspicious activity is not directly proportional to transaction size, with the \$5,000–\$9,999 range showing the strongest observed alert concentration.

Clients with structuring pattern

```
387 -- Q23: Clients with an unusually high volume of transactions clustered just under a round-number threshold $10,000
388 -- reporting line(classic structuring red flag) ?
```

Result Grid			
Filter Rows:			
Export:			
Wrap Cell Content: ☐			
	client_id	client_name	txns_near_threshold
▶	41	May, Jennings and Do...	3
	847	Johnson and Sons	3
	273	Marsh, Carson and Al...	3
	1332	Singleton, Simpson an...	3
	699	Dawson-Young	3
	1090	Murphy-Nolan	3
	717	Jones-Atkinson	3
	1067	Hansen, Dixon and Su...	3
	1323	Gregory, Fisher and J...	3
	1943	Johnson-McLaughlin	3
	961	Smith, Boyd and Myers	3
	693	Dean LLC	3
	282	Carey, Holland and A...	3
	649	Hicks and Sons	3
	1048	Pineda LLC	3
	1089	Johnston PLC	3
	1108	Cooper, Gray and Lewis	3
	1375	Daniel-Davies	3
	993	Lopez-Hudson	3
	75	Hunter, Green and Ga...	3

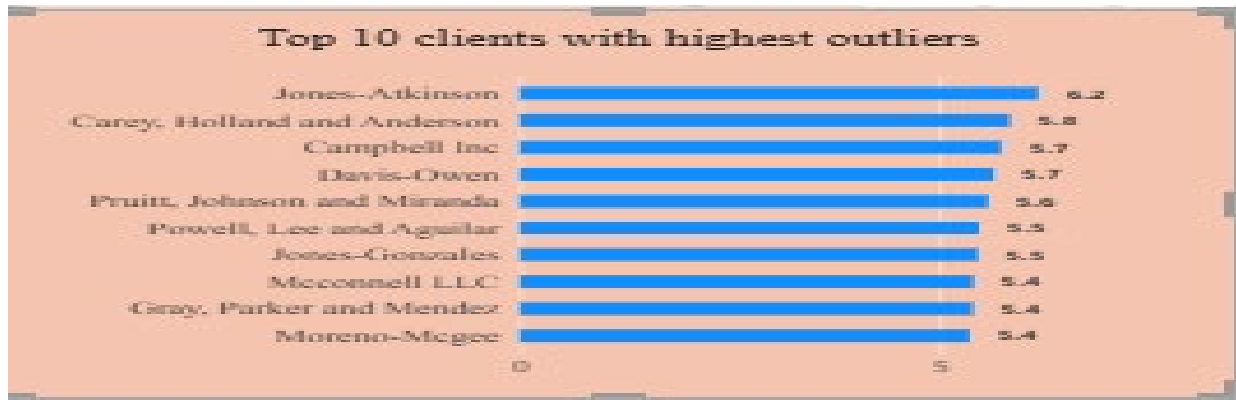
The round-number threshold analysis identifies multiple clients with three transactions clustered just below the \$10,000 threshold. The repeated near-threshold activity across multiple clients represents a potential structuring indicator and warrants further review of transaction timing, amounts, counterparties, and associated alert rules. The pattern should be treated as an investigation trigger rather than evidence of intentional structuring.

#OUTLIER ANALYSIS:

Clients with highest difference in their transaction amount size

```
398 -- Q24: Outlier transactions – which individual transactions sit furthest from
399 -- their OWN client's typical transaction size?
```

Result Grid							
Filter Rows:							
Export:							
Wrap Cell Content: ☐							
Fetch rows:							
	transaction_id	client_id	client_name	amount	avg_amount	stddev_amount	z_score
▶	47709	717	Jones-Atkinson	154936.76	7102.09	23912.09	6.18
	21003	282	Carey, Holland and A...	132049.01	7392.85	21343.23	5.84
	32454	1947	Campbell Inc	125574.23	7681.12	20535.69	5.74
	18143	1389	Davis-Owen	126091.69	6790.94	21111.06	5.65
	4712	784	Pruitt, Johnson and Mi...	93826.38	5895.38	15766.86	5.58
	34984	958	Powell, Lee and Aguilar	124217.96	7482.27	21309.87	5.48
	18798	1764	Jones-Gonzales	52742.54	3457.61	9025.87	5.46
	14984	1133	McConnell LLC	62434.11	4262.83	10760.59	5.41
	35231	1385	Gray, Parker and Men...	69619.5	4646.78	12034.21	5.4
	32512	292	Moreno-Mcgee	135920.5	8299.14	23798.88	5.36
	33568	1124	Chandler, Boyd and G...	88523.3	6207.37	15362.72	5.36
	33685	1991	Roman PLC	40477.87	3112.08	6982.74	5.35
	26403	875	Horton Group	110876.09	7555.12	19338.44	5.34
	2625	1060	Davis, Chang and Moore	96886.86	5952.38	17071.34	5.33
	34452	498	Miller-Baker	60633.57	4421.9	10637.88	5.28
	46101	303	Mcdonald and Sons	75458.24	4729.31	13385.54	5.28
	49763	980	Martin, Villarreal and S...	74481.14	5303	13233.42	5.23
	10502	894	Price, Ruiz and Douglas	44275.97	3766.41	7763.75	5.22
	15630	787	Nguyen PLC	28275.96	2535.35	4934.47	5.22
	32878	674	Sparks-Gray	103997.08	6220.06	18784.23	5.21



The outlier analysis identifies multiple transactions that are substantially larger than the respective clients' historical transaction patterns. The highest outlier has a z-score of 6.18, with a transaction amount of approximately \$154.9 K compared with an average client transaction size of approximately \$7.1 K. All displayed outliers have z-scores above 5, indicating significant deviations from normal client behaviour. These transactions represent important behavioural-anomaly alerts for further investigation, although a high z-score alone does not establish suspicious or illicit activity.

#CLIENT PRIORITIZATION / CASE QUEUE ANALYSIS

Clients to be monitored on Priority:

```
402 -- Q25: Find priority list of customers for the compliance team's review queue which blends
403 -- suspicious volume, alert intensity, and dollar exposure?
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	client_id	client_name	sector	count	overall_ris	total_trans	suspicious_tran	total_alert_count	total_value	priority_score
▶	1323	Gregory, Fisher and J...	Retail	SY	High	28	28	40	106205.99	124.11
	1849	Jones-Bailey	Retail	KP	High	29	29	36	154854.19	123.15
	1785	Alexander-Johnson	Energy/Oil	US	High	30	28	36	327757.14	120.33
	1433	Lozano, Oconnell and ...	Casino/Gambl...	VE	High	32	26	38	110998.64	116.11
	1879	Dixon-Butler	Energy/Oil	AU	High	28	24	36	177733.95	108.18
	1788	Merritt LLC	Import/Export	VE	High	28	23	35	117698.39	104.12
	1771	Foley, Johnson and L...	Defense/Arms	AE	High	29	23	32	260902.75	101.26
	562	Miller-Williams	NGO/Charity	SG	High	29	23	31	327951.8	100.33
	1787	Mann, Jones and Gon...	Crypto Excha...	SD	High	29	23	31	239662.05	100.24
	911	Macias-Riggs	Crypto Excha...	VE	High	30	22	33	81465.57	99.08
	1283	Fowler, Smith and Bis...	Defense/Arms	VE	High	28	22	31	377581.33	97.38
	585	George and Sons	Import/Export	JP	High	22	22	31	229634.8	97.23
	1216	Jackson, Smith and Ri...	NGO/Charity	CN	High	26	22	31	194609.68	97.19
	665	Lopez, Bell and Spears	Energy/Oil	LB	High	29	21	34	174198.46	97.17
	1206	Campbell PLC	Casino/Gambl...	JP	High	30	23	28	112529.76	97.11
	3	Phillips-Hanson	NGO/Charity	AE	High	25	22	30	61924.06	96.06
	944	Suarez Group	Defense/Arms	AU	High	25	21	31	287304.47	94.29
	751	Choi-Short	Import/Export	RU	High	29	22	28	110100.8	94.11
	714	Gallegos and Sons	Import/Export	FR	High	22	22	28	59798.89	94.06
	1779	Hawkins Ltd	Real Estate	LB	High	21	21	30	146260.84	93.15

The compliance priority analysis identifies a group of High-risk clients with elevated combinations of suspicious activity, alert intensity, transaction volume, and financial exposure. Jones-Bailey and Alexander-Johnson have the highest visible priority scores at approximately 123, while several Energy/Oil and other sector clients also appear prominently in the review queue. The ranking demonstrates that transaction value alone does not determine review priority; clients with lower transaction values can receive higher priority when they exhibit greater suspicious or alert activity. The priority score therefore provides a practical mechanism for organizing enhanced review within the High-risk population

New Clients with highest suspicious rate:

418 | -- Q26: Find clients with comparatively few transactions, but a disproportionate share has already flagged ?

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	client_id	client_name	total_txns	suspicious_txns	suspicious_rate_pct
▶	822	Snyder PLC	12	12	100.0
	626	Smith-Santos	12	4	33.3
	1791	Cowan LLC	10	2	20.0

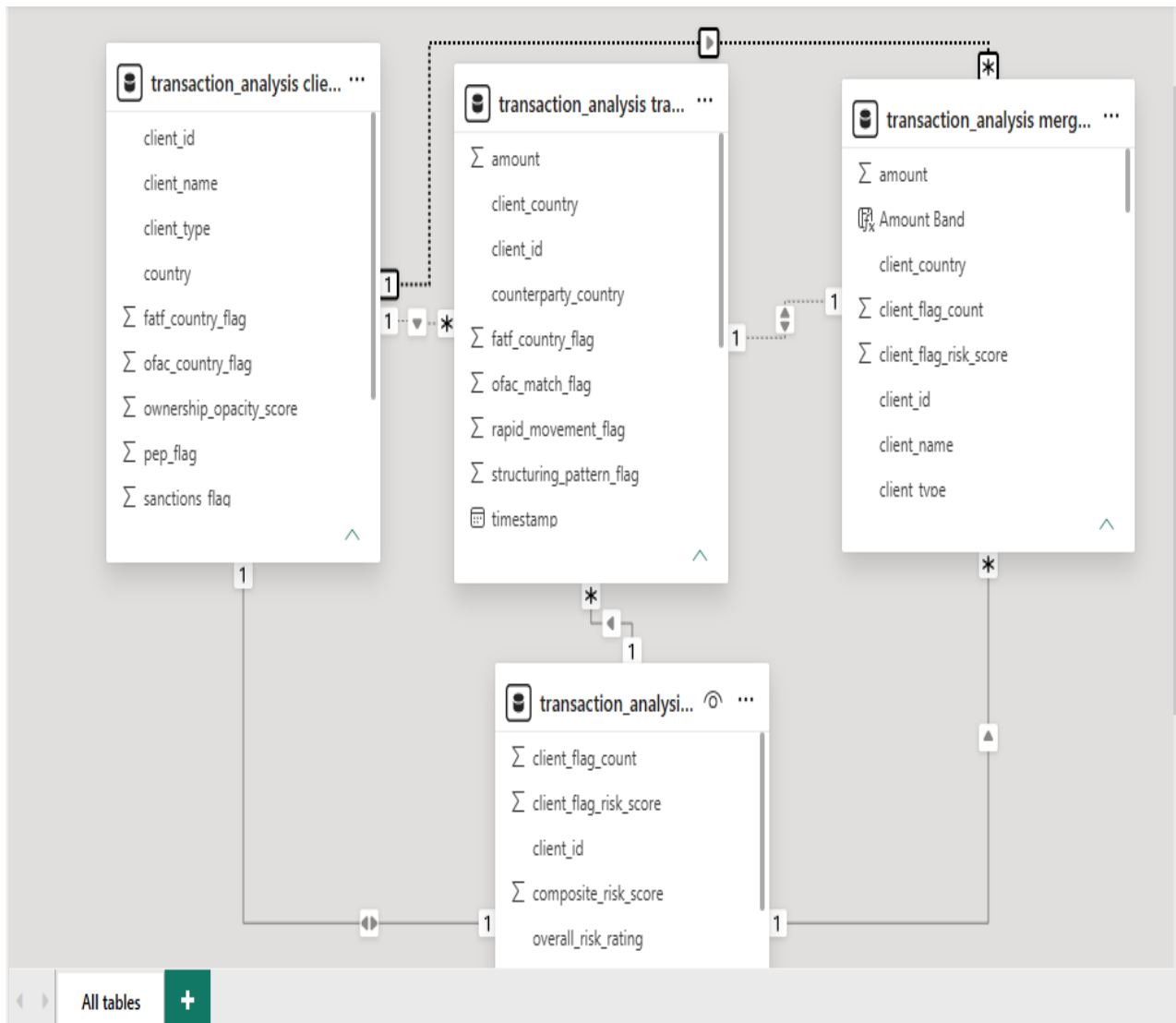
The analysis identifies several clients with relatively low transaction volumes but unusually high suspicious transaction rates. Snyder PLC is the most notable example, with all 12 of its transactions flagged as suspicious, resulting in a 100% suspicious rate. Smith-Santos and Cowan LLC also show elevated rates of 33.3% and 20.0%, respectively. These clients represent potential monitoring priorities because suspicious activity forms a disproportionately large share of their transaction activity. However, the relatively small transaction populations mean that the rates should be interpreted alongside absolute suspicious counts and investigated at the transaction and alert-rule level.

Data Analysis using Power BI:

Connected MYSQL Database named “transaction_analysis” to Power BI for further Analysis and Data Visualization by Creating charts.

Data Modelling:

Relationship is created between tables to connect the data tables to create a logical semantic model. Power BI can use related tables in DAX calculations and visuals.



Required custom measures and calculated columns are created using DAX for business logic. Measures are better for interactive numeric calculations. Calculated columns are evaluated row by row and are appropriate for fields used in axes, legends, rows, columns, slicers, and filters.

Data Visualization:

Interactive visuals are Built with dynamic charts, maps, cards, and tables that cross-filter related data. It uses visual tiles to highlight key business metrics and tell a data-driven story.

AML TRANSACTION MONITORING ANALYSIS DASHBOARD

Client Type

- ☐ Corporate
- ☐ Financial Institution
- ☐ Individual
- ☐ NGO

2,000
unique_clients

50000
Total_txns

\$20,98,06,892.97
Total_txn_value

8.98%
suspicious_txn_...

4489
suspicious_txn_co
unt

Mean_txn_value
4,196.14

Median_txn_value
1,582.19

Mean-to-Median ratio
2.65

High Risk Clients
1218

High_risk_client_tran...
36,370.00

Transaction Type

ACH

SWIFT

CHECK

WIRE

Unique Clients by client type



Total_txns By Transaction Type



Transaction Volume and Suspicious Activity by Month



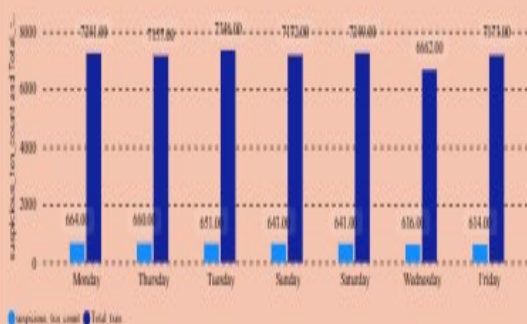
Transaction month

- ☐ 2025-07
- ☐ 2025-08
- ☐ 2025-09

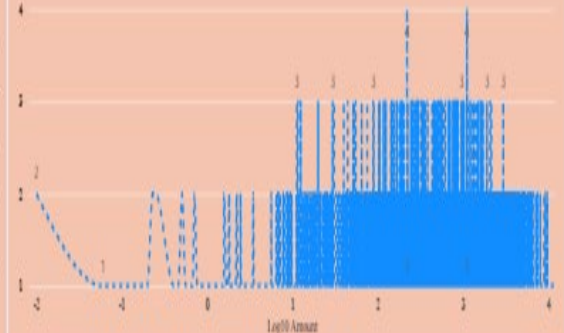
Day of week

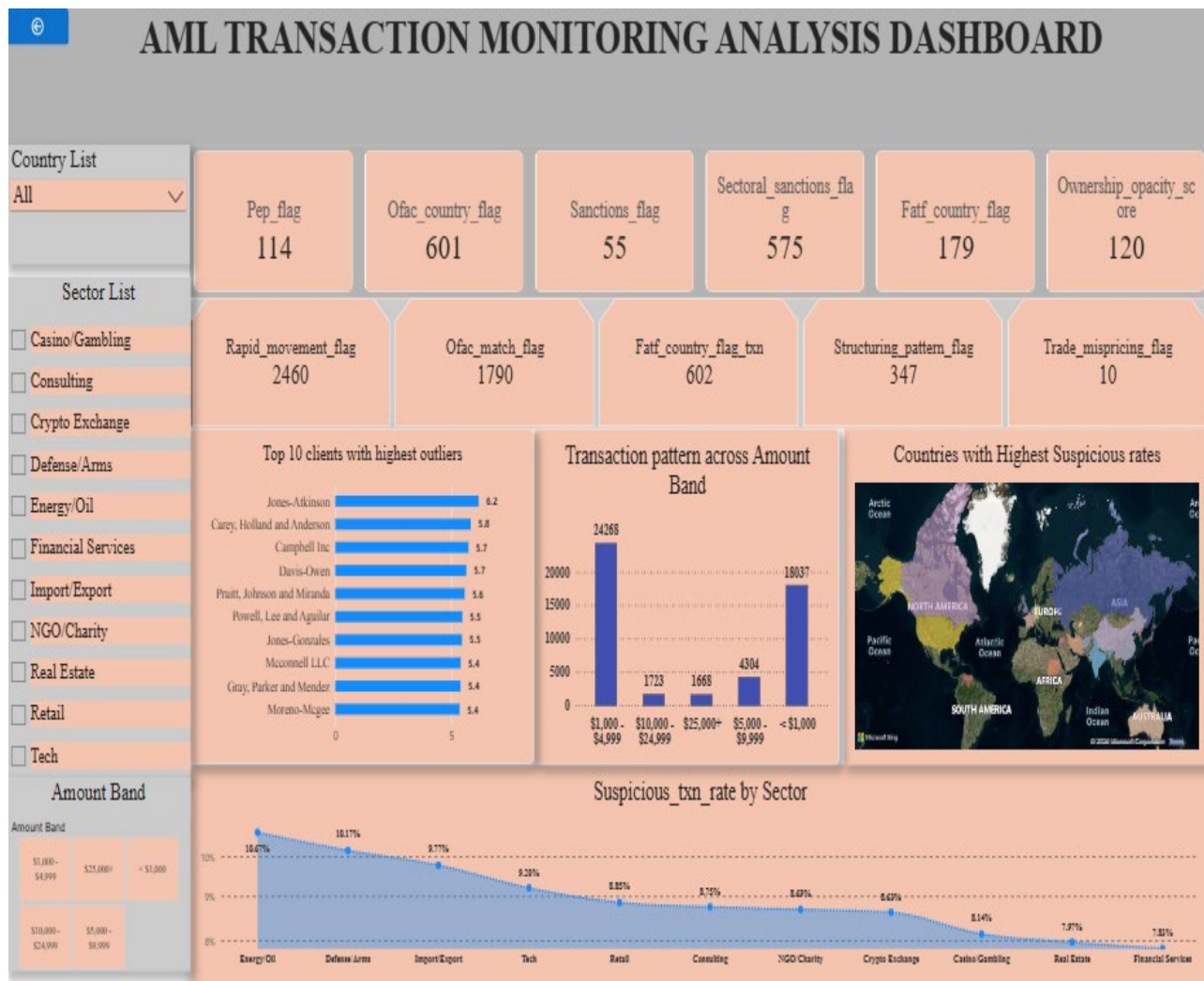
All

Total Transaction Vs Suspicious_Txn Daily Activity



Transaction amount Distribution





Conclusion:

The AML/KYC analysis demonstrates that suspicious transaction activity is driven by a combination of sector exposure, country risk, transaction characteristics, monitoring-rule triggers, and deviations from individual client behaviour. High-risk sectors such as Energy/Oil and Defence/Arms recorded elevated suspicious rates, while certain country populations showed concentrated FATF and OFAC risk indicators. Transaction monitoring workload was primarily concentrated in Rapid Movement and OFAC Match alerts, while the low co-occurrence between Structuring and Rapid Movement suggests complementary rule coverage. The \$8,000–\$9,999 transaction band and several client-level behavioural outliers also emerged as notable areas for investigation. Overall, the findings support a risk-based monitoring approach that combines static KYC risk indicators with transaction-level and behavioural analysis rather than relying on any single risk factor.